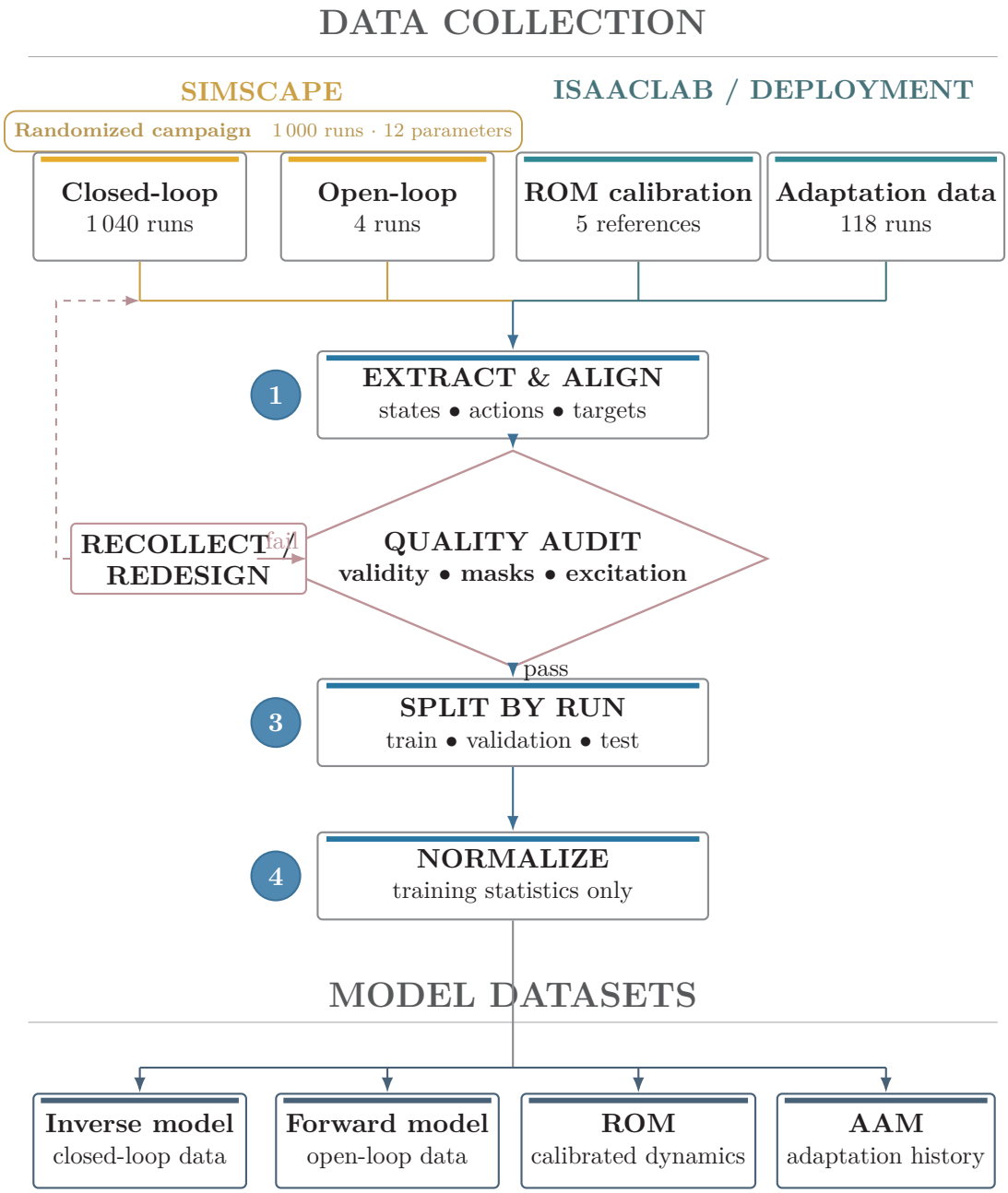


Figure 4.1: Data-preparation workflow. Two Simscape collections and two IsaacLab collections pass through common extraction, audit, run-level splitting and training-statistics normalization stages. The randomized Simscape campaign is provenance metadata rather than a pipeline stage. A failed audit returns the experiment to collection; accepted material forms the inverse, forward, ROM and AAM datasets.